Q14/

Find the value of $x^3+\frac{1}{x^3}$ when $x+\frac{1}{x}=7$

sol>

$x^3+\frac{1}{x^3}=\,?\quad ? \qquad x+\frac{1}{x}=7$

so

$x+\frac{1}{x}=7$

Taking cube on b/s:-

$\left(\underset{a}{x}+\underset{b}{\frac{1}{x}}\right)^3=(7)^3$

$(a+b)^3=a^3+b^3+3ab(a+b)$

$x^3+\frac{1}{x^3}+3\cancel{(x)}\frac{1}{\cancel{(x)}}\left(x+\frac{1}{x}\right)=(7)^3$

$x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)=(7)^3=\left(x^3+\frac{1}{x^3}\right)+3(7)=343$

$\cancel{x^3+\frac{1}{x^3}=\left(\frac{343}{3}\right)-x-\frac{1}{x}}\quad \left(x^3+\frac{1}{x^3}\right)=343-21$

$\left(x^3+\frac{1}{x^3}\right)=322$

Ans

Q15/ same just different symbol:
we will use

ii) $(a-b)^3=a^3-b^3-3ab(a-b)$

$8x^6 - 12x^4y^2 +$

$\Rightarrow$

$(a+b)(a^2-ab+b^2) = a^3+b^3$.

compare the given product with that pattern.

$(2x^2+3y^2)(4x^4-6x^2y^2+9y^4)$

set $a = 2x^2$ and $b = 3y^2$, then

$a^2 = (2x^2)^2 = 4x^4$

$-(a \times b) = 2x^2 \times 3y^2 = -6x^2y^2$

$+b^2 = 3y^2 = (3y)^2 = 9y^4$

So the second factor is exactly

(a^2-ab+b^2)

By the identity :-

$(2x^2+3y^2)(4x^4-6x^2y^2+9y^4) = (2x^2)^3 + (3y^2)^3$

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Q15, same just different symbol.

we will use:

ii, $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$

Now compute cubes:

$(2x^2)^3 = 2^3(x^2)^3 = 8x^6$, $(3)^3 \cdot (y^2)^3$

$= 8x^6 + 27y^6$ Ans

Now you can verify by expanding L.H.S and collect like terms then it will give you same result.

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